Classical Mechanics

Lecture 2: Initial Data, Conservation, and the Action

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Chapter 1

Initial Data, Conservation, and the Action

The laws of mechanics can be stated in two remarkably different ways. One description is local: specify the state now, use a differential equation, and advance the motion from one instant to the next. The other description treats an entire trajectory as a single object and selects the physical path by a stationary principle. Our purpose is to understand why both descriptions carry the same physics.

We shall approach the global principle slowly. First we return to initial data and the order of a differential equation. We then review energy and momentum, not because their elementary Newtonian derivations are the deepest proofs of those laws, but because they reveal structures that survive far beyond the derivations. Only after that preparation shall we pass from ordinary functions to functionals and uncover the action.

1.1 What the Order of a Law Demands

Begin in one dimension with a deliberately false law,

$$F = m\dot{x} = m\frac{dx}{dt}. \tag{1.1}$$

It says that a force is needed merely to sustain motion: remove the force and the body stops at once. This is not Newtonian physics. Its value is precisely that it lets us see what a first-order equation would demand of its initial data.

Suppose the force is prescribed by position,

$$F(x) = m\dot{x}. \tag{1.2}$$

If the position is known at one instant, then $F(x)$ is known, and the law immediately supplies the velocity,

$$\dot{x} = \frac{F(x)}{m}. \tag{1.3}$$

The same datum fixes the acceleration. Differentiating the force along the motion gives

$$\frac{dF}{dt} = \frac{dF}{dx}\dot{x}, \tag{1.4}$$

and differentiating Eq. (1.2) gives

$$m\ddot{x} = \frac{dF}{dx}\dot{x}. \tag{1.5}$$

Differentiate once more and use the product and chain rules:

$$m\,\dddot{x} = \frac{d}{dt}\left(F'(x)\dot{x}\right) \tag{1.6}$$

$$= F''(x)\dot{x}^{\,2} + F'(x)\ddot{x}. \tag{1.7}$$

Thus position determines velocity, velocity and position determine acceleration, and the same chain continues through jerk and every higher time derivative. In this first-order world, one number $x(t_0)$ is enough to start the motion.

Throughout this calculation the mass is constant. A visible subsystem may seem to lose mass if a piece falls away, but then the discarded piece has simply been omitted from the description. For the complete closed system, the mass carried by the calculation has not disappeared.

Now replace the false law by Newton's law,

$$F(x) = m\ddot{x}. \tag{1.8}$$

Position still determines force and therefore acceleration, but it does not determine velocity. Two bodies can pass through the same point with different velocities and subsequently follow different motions. We must supply

$$x(t_0) = x_0, \qquad \dot{x}(t_0) = v_0. \tag{1.9}$$

Once these two data are known, Newton's equation supplies $\ddot{x}$, and successive differentiations supply the rest. The need for position and velocity is not an arbitrary convention: it is the signature of a second-order law.

For many particles, the statement simply expands. We must specify every particle coordinate, every corresponding velocity component, and the force law, which may depend on the relative positions of all the particles.

Question from the audience. Why does Newton's law require both position and velocity, whereas the false first-order law requires only position? [a]

Response. The distinction is the order of the equation. In $F(x) = m\dot{x}$, the position determines the first derivative directly. In $F(x) = m\ddot{x}$, it determines only the second derivative. The first derivative remains independent and must be included in the initial state.

[a] Lecture timestamp: 00:09:14–00:11:57.

1.2 Energy and Conservative Forces

Energy conservation is more general than the elementary argument we are about to give. It remains meaningful in electrodynamics, special and general relativity, and quantum mechanics. Here we ask the narrower question: under what conditions does ordinary Newtonian particle mechanics conserve $T + U$?

The qualification matters. A friction force appears to destroy mechanical energy if the resulting heat and the microscopic degrees of freedom of the environment are ignored. An even sharper counterexample is a tangential force that continually pushes a constrained particle around a circle. After one revolution the particle returns to the same position with a larger speed. Any single-valued potential has returned to its original value, while the kinetic energy has increased. Such a force cannot be written as the gradient of an ordinary potential.

For the class of forces considered here, there is a potential energy $U(\mathbf{x})$ such that

$$F_i = -\frac{\partial U}{\partial x_i}, \qquad \mathbf{F} = -\boldsymbol{\nabla} U. \tag{1.10}$$

The minus sign says that the force points toward decreasing potential energy.

Define

$$T = \frac{1}{2} m \mathbf{v}^2 = \frac{1}{2} m \sum_i v_i^2, \qquad E = T + U. \tag{1.11}$$

The board develops the proof in the useful order: write the two energies, differentiate them separately, and only then combine them.

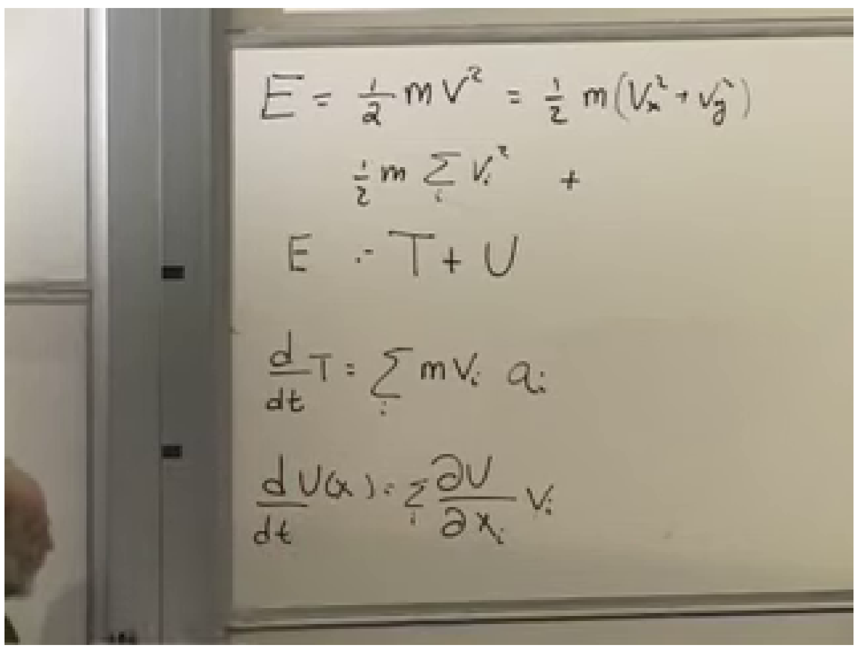

Figure 1.1: The energy calculation at 00:22:24. The upper handwritten label is read as T: the formula, transcript, and later identity $E = T + U$ fix that interpretation.

Lecture frame: 00:22:24.

For the kinetic energy,

$$\frac{dT}{dt} = \frac{d}{dt}\left(\frac{1}{2} m \sum_i v_i^2\right) \tag{1.12}$$

$$= \sum_i m v_i \frac{dv_i}{dt} = \sum_i m v_i a_i. \tag{1.13}$$

The potential changes because the particle moves through space. The multivariable chain rule gives

$$\frac{dU}{dt} = \sum_i \frac{\partial U}{\partial x_i}\frac{dx_i}{dt} \tag{1.14}$$

$$= \sum_i \frac{\partial U}{\partial x_i} v_i. \tag{1.15}$$

Adding the two derivatives,

$$\frac{dE}{dt} = \sum_i v_i \left(m a_i + \frac{\partial U}{\partial x_i} \right) \tag{1.16}$$

$$= \sum_i v_i (F_i - F_i) = 0. \tag{1.17}$$

The first F_i is Newton's ma_i; the second comes from $\partial_i U = -F_i$. The cancellation is exact.

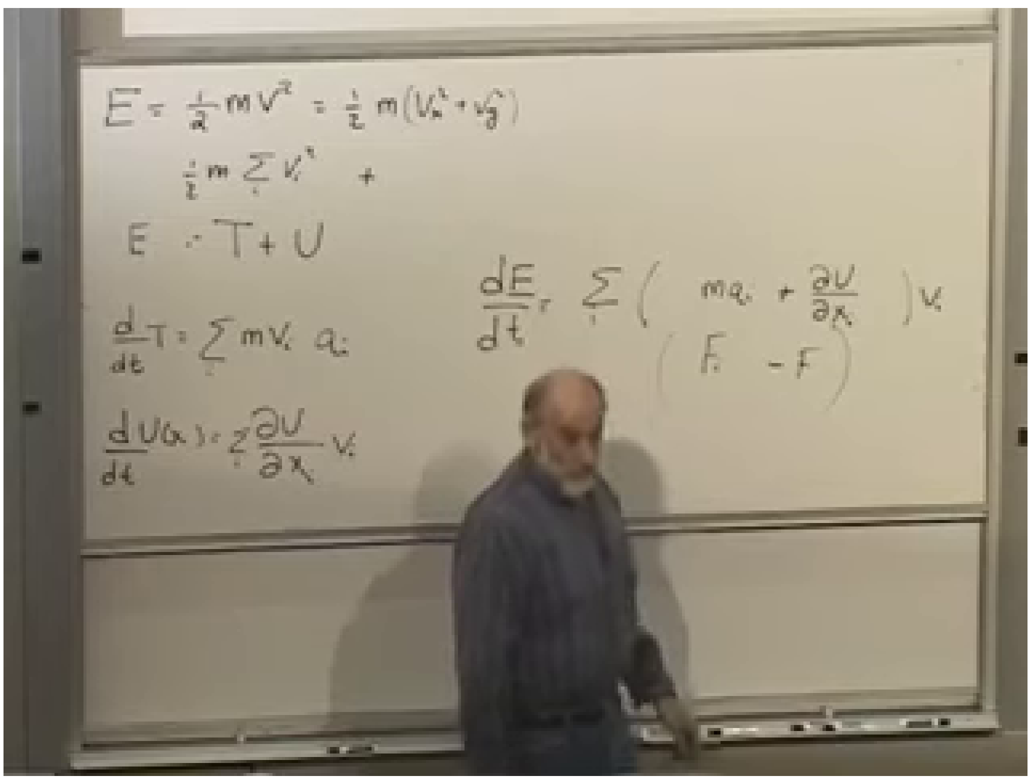

Figure 1.2: The completed energy derivative at 00:24:01. The right-hand bracket is the decisive $F_i - F_i$ cancellation.

Lecture frame: 00:24:01.

Nothing in the argument restricts i to the three directions of one particle. It may range over all coordinates of all particles. If each generalized force is the negative derivative of one common potential and the potential has no explicit time dependence, the total energy of the complete system is conserved.

Question from the audience. Why denote potential energy by U rather than the common letter V?[a]

Response. Either notation is possible. Here U avoids collisions: V is often used for electric voltage and v already denotes velocity. The symbol changes nothing in the physics.

[a]Lecture timestamp: 00:18:13–00:18:36.

1.3 Momentum and a Closed System

Momentum enters by rewriting the same Newtonian equation. With constant mass,

$$F_i = m\frac{dv_i}{dt} = \frac{d}{dt}(mv_i). \tag{1.18}$$

Define

$$\mathbf{p} = m\mathbf{v}. \tag{1.19}$$

Then Newton's law becomes

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}. \tag{1.20}$$

In the absence of force, $\dot{\mathbf{p}} = 0$, so a free particle carries a constant momentum. Intergalactic space is not perfectly empty, but it gives a good approximation to this ideal.

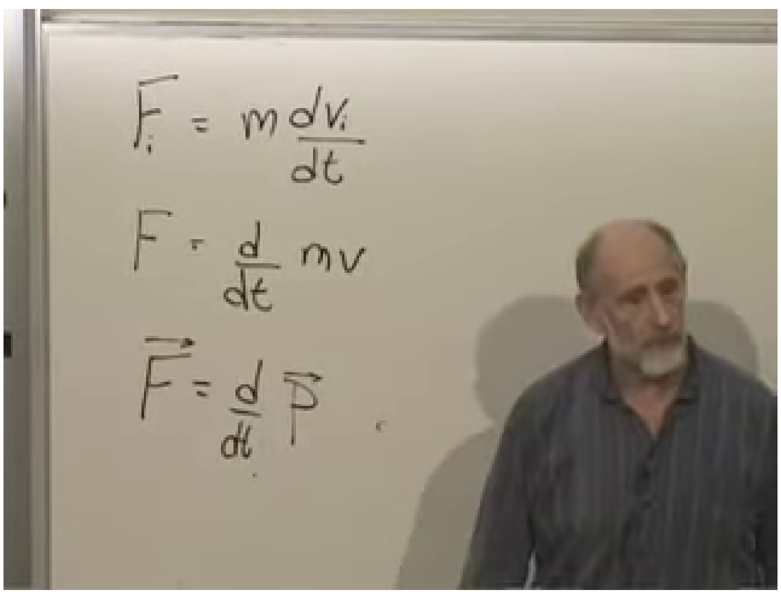

Figure 1.3: The component, scalar, and vector forms of Newton's momentum law at 00:29:00.
Lecture frame: 00:29:00.

This discussion uses a specifically Newtonian picture of time. Clocks can be synchronized everywhere, all observers agree on one universal time, and time flows uniformly. We are also considering force laws with no prescribed external time dependence. These assumptions will not survive unchanged in relativity, so they should be stated rather than hidden.

Now consider three interacting particles. Let $\mathbf{F}_{ab}$ mean the force on particle a due to particle b. Internal forces give

$$\dot{\mathbf{p}}_1 = \mathbf{F}_{12} + \mathbf{F}_{13}, \tag{1.21}$$
$$\dot{\mathbf{p}}_2 = \mathbf{F}_{21} + \mathbf{F}_{23}, \tag{1.22}$$
$$\dot{\mathbf{p}}_3 = \mathbf{F}_{31} + \mathbf{F}_{32}. \tag{1.23}$$

Newton's action–reaction law states

$$\mathbf{F}_{ab} = -\mathbf{F}_{ba}. \tag{1.24}$$

The equal-and-opposite forces lie along the line joining the pair in the elementary central-force picture, although only their cancellation is needed here.

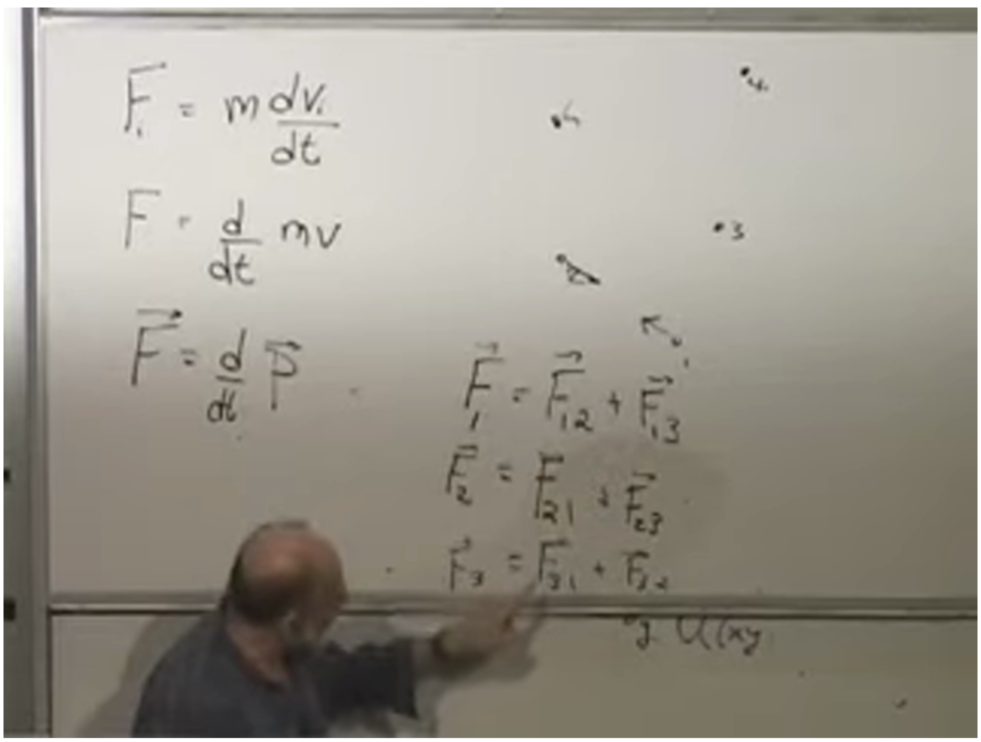

Figure 1.4: The board at 00:32:58, after the three force balances have been assembled. The frame is used for their layout; the exact subscripts are typeset in Eq. (1.23).

Lecture frame: 00:32:58.

Adding the three equations gives

$$\begin{aligned}\frac{d}{dt}(\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3) &= (\mathbf{F}_{12} + \mathbf{F}_{21}) + (\mathbf{F}_{13} + \mathbf{F}_{31}) \\ &\quad + (\mathbf{F}_{23} + \mathbf{F}_{32}) \\ &= 0. \end{aligned} \tag{1.25}$$

Therefore

$$\frac{d\mathbf{P}}{dt} = 0, \qquad \mathbf{P} = \sum_a \mathbf{p}_a. \tag{1.26}$$

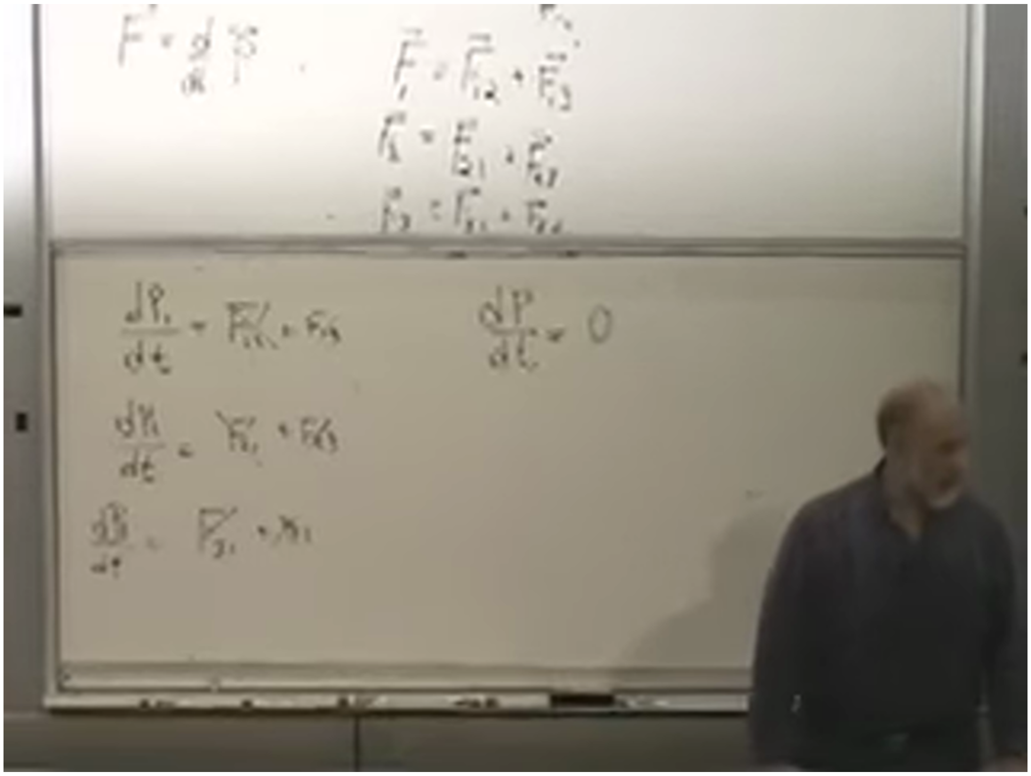

Figure 1.5: The completed cancellation board at 00:35:20. Each internal force appears once with each orientation, leaving $\dot{\mathbf{P}} = 0$.

Lecture frame: 00:35:20.

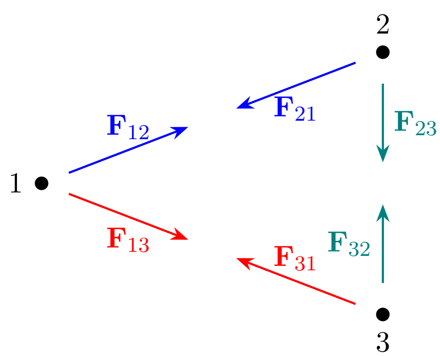

Figure 1.6: Pairwise internal forces cancel when the momentum equations for a closed system are summed.

A closed system is one for which every relevant force source has been included. If an external force is present, it does not acquire a partner inside the sum, and the result becomes $\dot{\mathbf{P}} = \mathbf{F}_{\text{ext}}$.

Question from the audience. Are action and reaction instantaneous? [a]

Response. They are instantaneous in Newtonian mechanics, where the speed at which influences propagate is effectively infinite. Real influences do not propagate instantaneously; special relativity forbids that. Yet momentum conservation survives. The Newtonian derivation is therefore less general than the conservation law it reveals.

[a]Lecture timestamp: 00:36:15–00:37:12.

Momentum conservation applies to radiation pressure, quantum systems, and relativistic physics, even though the simple pair-force proof cannot be carried unchanged into every one of those settings. There are three momentum laws, one for each spatial component, and one energy law. In special relativity these four quantities belong together as energy–momentum. This is a first hint that the conserved quantities have a deeper origin than the present calculations.

Question from the audience. Why does momentum seem to provide more conservation laws than energy?[a]

Response. Momentum has one conserved component for each independent direction of space, whereas energy is one scalar associated with evolution in time. In three spatial dimensions that gives three momentum components and one energy. Relativity later unifies the four rather than treating them as an accidental list.

[a]Lecture timestamp: 00:37:37–00:38:24.

1.4 Stationarity in Ordinary Calculus

We now prepare for what is really the central law of the course: the principle of least action. The required mathematics begins with an ordinary function of one variable, $f(y)$. At a smooth local minimum,

$$\frac{df}{dy} = 0. \tag{1.27}$$

The derivative also vanishes at a smooth maximum. It can even vanish at a flat inflection point, such as the origin of $f(y) = y^3$. The first-derivative condition therefore identifies a *stationary point*; the second derivative and higher information distinguish the possibilities.

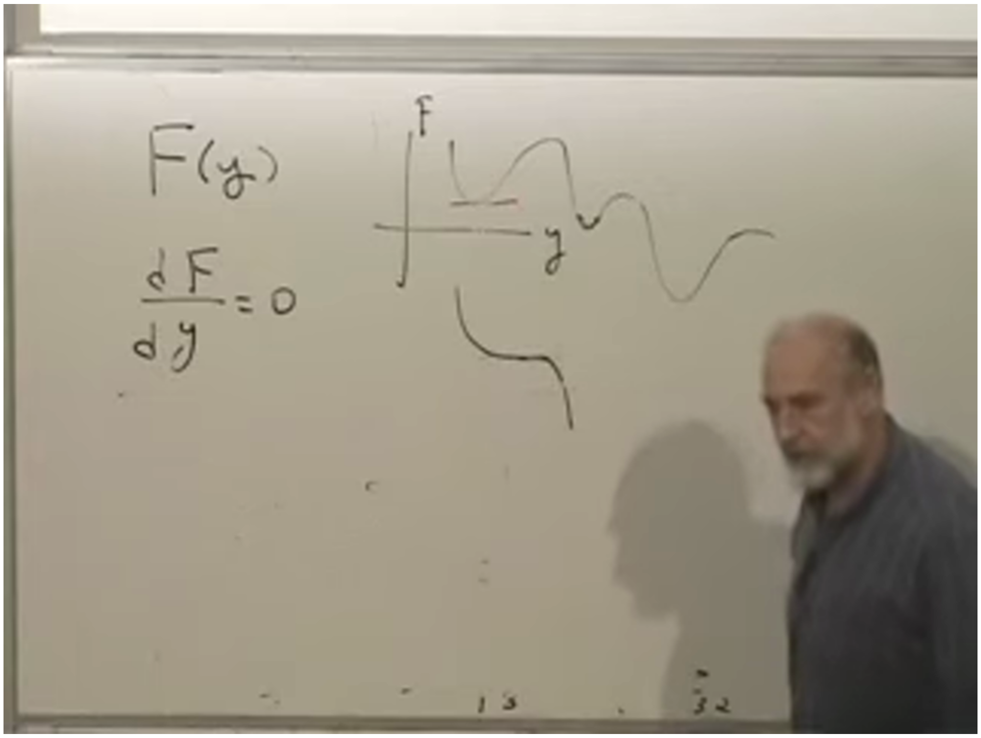

Figure 1.7: Minimum, maximum, and flat inflection behavior on the board at 00:43:20. All are stationary to first order.

Lecture frame: 00:43:20.

For a function of many variables, stationarity must hold along every coordinate direction:

$$\frac{\partial f}{\partial x_i} = 0 \qquad \text{for every } i. \tag{1.28}$$

Geometrically, at the bottom of a bowl there is no downhill first-order direction. The same equations also find maxima and saddles, so “stationary” remains the accurate word.

As an example, take

$$f(x,y) = x^2 + y^2 + xy + 3x + 2y + 7. \tag{1.29}$$

The stationarity equations are

$$\frac{\partial f}{\partial x} = 2x + y + 3 = 0, \tag{1.30}$$

$$\frac{\partial f}{\partial y} = x + 2y + 2 = 0. \tag{1.31}$$

Solving the linear system gives

$$x = -\frac{4}{3}, \qquad y = -\frac{1}{3}. \tag{1.32}$$

The constant 7 never enters the equations because a vertical shift changes the value of f, not the location of its stationary point.

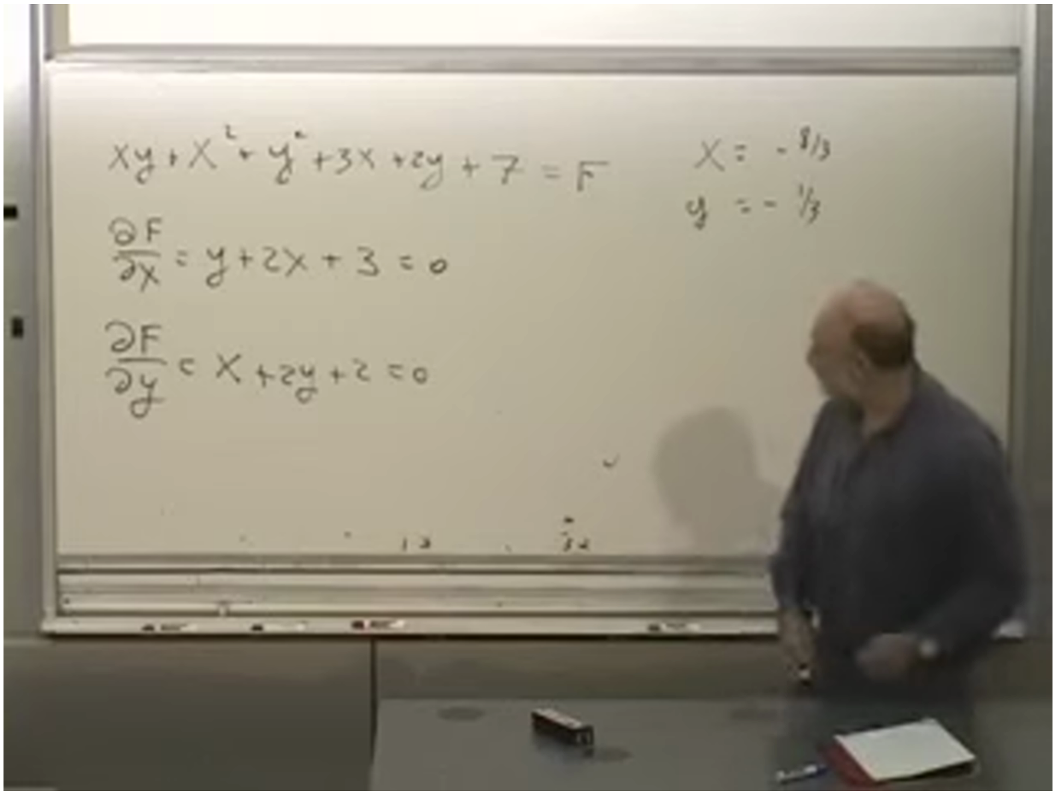

Figure 1.8: The two-variable example at 00:51:10. The displayed derivative equations are correct; the board's $x = -8/3$ is an arithmetic slip. Solving those equations gives Eq. (1.32).

Lecture frame: 00:51:10.

For n variables there are n first-order equations. They may have no solution, one solution, or several solutions. A monotone function need not have a stationary point, while a landscape with several valleys can have many.

Question from the audience. Is the physical principle really one of least action, or of stationary action?[a]

Response. Stationary action is the precise statement. The first-order change of the action vanishes; that does not by itself prove a minimum. In many elementary mechanical problems the physical path is locally minimizing, so "least action" is useful language, but the distinction should not be forgotten. Velocity-dependent interactions are one reason not to build the theory around a naive minimum test.

[a] Lecture timestamp: 00:42:05–00:46:39.

1.5 Generalized Trajectories

The basic mathematical problem of mechanics is to determine the trajectory of a system from suitable data. For 10^{23} particles, no one attempts to write every trajectory in practice; statistical or approximate descriptions are used instead. Nevertheless, the exact classical question remains well defined.

Describe a system by generalized coordinates

$$q_1, q_2, \ldots, q_N. \tag{1.33}$$

They may be Cartesian positions, angles, orientations, latitude and longitude, or any other independent variables that locate the system in configuration space. A single particle in three-dimensional space has three such coordinates; two particles have six; n unconstrained particles have $3n$.

The motion is the whole collection

$$\mathbf{q}(t) = \big(q_1(t), q_2(t), \ldots, q_N(t)\big). \tag{1.34}$$

This curve through configuration space is a generalized trajectory. To begin a Newtonian prediction, we specify both $\mathbf{q}(t_0)$ and $\dot{\mathbf{q}}(t_0)$. Newton's equations then act locally: knowing where the system is and how it is moving tells us its acceleration, and hence how to advance to the neighboring point of the path.

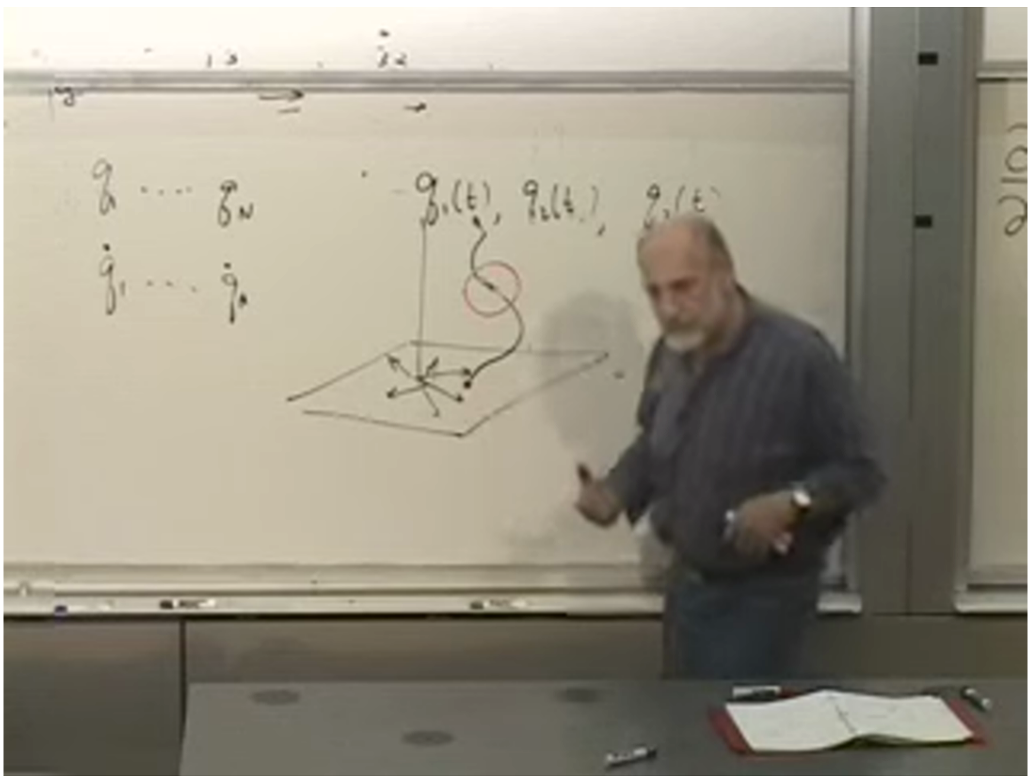

Figure 1.9: The generalized trajectory at 01:00:20. The horizontal plane stands for the many-dimensional configuration coordinates; the vertical direction represents time.

Lecture frame: 01:00:20.

There is another way to pose the problem. Instead of giving position and velocity at one time, hold the configurations fixed at two times,

$$\mathbf{q}(t_0) = \mathbf{q}_0, \qquad \mathbf{q}(t_1) = \mathbf{q}_1, \tag{1.35}$$

and ask which trajectory connects them. In the elementary well-posed cases being considered, this supplies the same amount of boundary data as position and velocity at one time. More complicated systems or long time intervals can admit several classical paths, but each stationary path still satisfies the same equations of motion.

The contrast is conceptual. Newton's differential equation gives a local rule along the curve. A variational principle assigns one number to the whole curve and asks which admissible curve makes that number stationary.

1.6 From Functions to Functionals

Take two fixed points in the xy-plane and describe a connecting curve by $y(x)$. The shortest connection is plainly a straight line, but we want the mathematical question that produces that answer.

For a small segment,

$$ds^2 = dx^2 + dy^2, \tag{1.36}$$

and therefore

$$ds = dx\sqrt{1 + \left(\frac{dy}{dx}\right)^2}. \tag{1.37}$$

Adding all the little segments gives the length

$$\mathcal{S}[y] = \int_{x_1}^{x_2} \sqrt{1 + (y'(x))^2}\, dx. \tag{1.38}$$

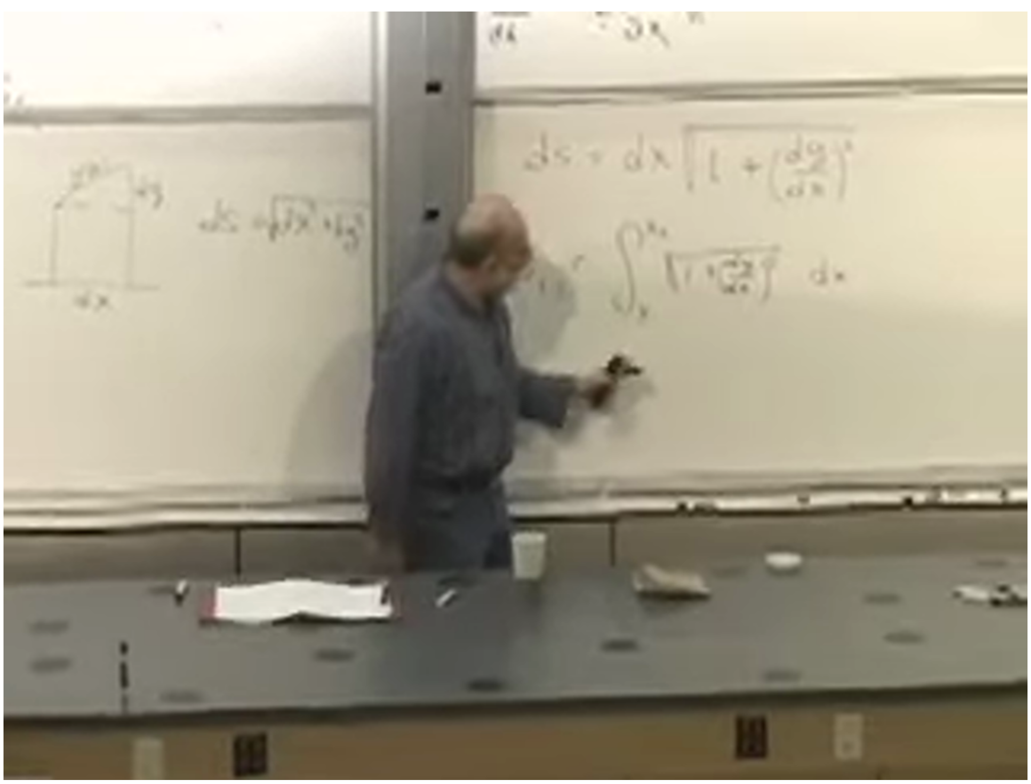

Figure 1.10: The curve, its dx-dy triangle, and the arc-length integral at 01:12:00.

Lecture frame: 01:12:00.

The input to $\mathcal{S}$ is not one number or a finite list of numbers. It is the entire function $y(x)$, and the output is a number. Such an object is a *functional*.

There are complementary global and local views. Globally, compare every curve between the endpoints and select a shortest one. Locally, break the curve into tiny pieces and focus on a point between two nearby pinned neighbors. Moving that point until the two small segments line up says simply: continue straight ahead. Applying the same local condition everywhere gives the straight line. The global extremization and the local differential equation encode the same geometry.

One tempting shortcut fails. Since the integrand in Eq. (1.38) is smallest when $y' = 0$, why not set $y' = 0$ at every point? Because a horizontal line will not generally meet the fixed endpoints. The admissible curves are constrained globally. Minimizing an integrand point by point while ignoring its boundary conditions solves a different problem.

1.6.1 Fermat's Principle

Optics supplies a more revealing functional. Suppose a layered isotropic medium has a light speed $c(y)$ that varies with height but not with direction at a fixed point. Fermat's principle says that the physical ray makes the travel time stationary. Locally,

$$c(y) = \frac{ds}{dt}, \qquad dt = \frac{ds}{c(y)}. \tag{1.39}$$

Using Eq. (1.37),

$$dt = \frac{\sqrt{1 + (y')^2}}{c(y)}\, dx, \tag{1.40}$$

so the total time is the functional

$$\mathcal{T}[y] = \int_{x_1}^{x_2} \frac{\sqrt{1 + \left(y'(x)\right)^2}}{c(y)}\, dx. \tag{1.41}$$

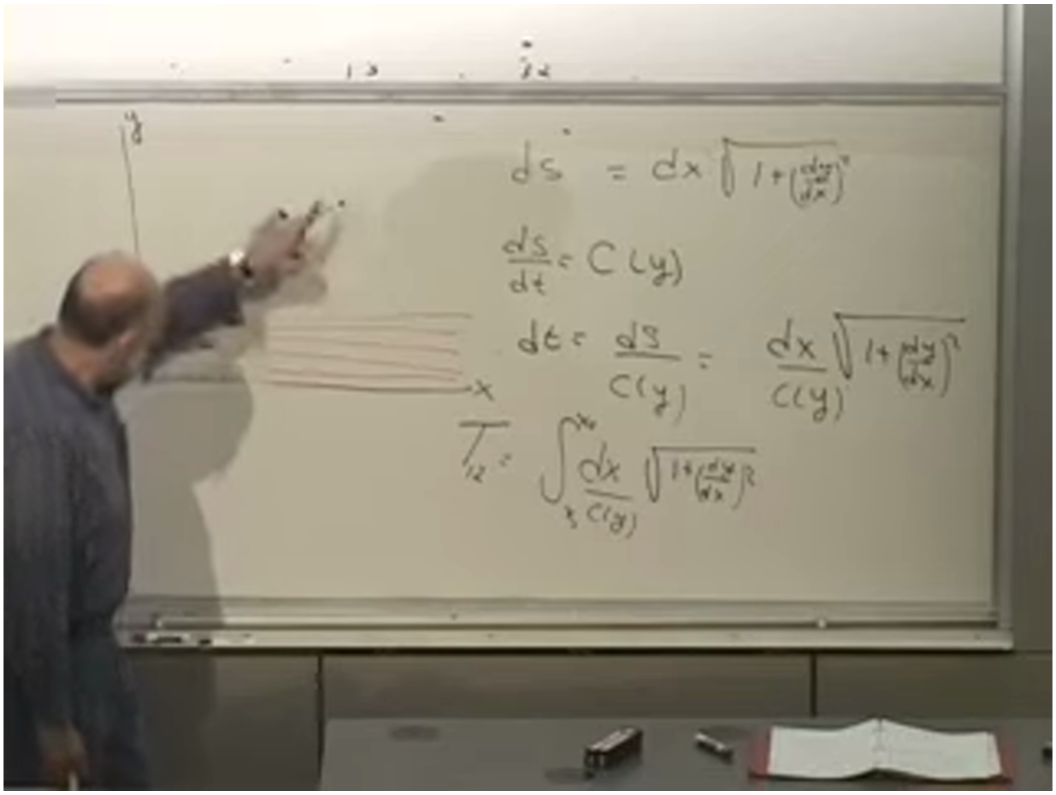

Figure 1.11: The layered-medium ray and completed least-time functional at 01:23:10. The ray bends toward the faster region while still advancing toward its endpoint.

Lecture frame: 01:23:10.

If c is constant, minimizing time is the same as minimizing distance, and the ray is straight. If light is faster higher in the medium, the best path can curve upward to spend more of its journey in the faster region. It cannot gain indefinitely by going straight up and then straight across: the vertical detour costs time without advancing toward the destination. The physical path is a compromise selected by the whole integral.

1.6.2 The Hanging Chain

A chain of fixed length suspended from two points supplies another variational problem. Gravity drives each small mass element downward, while the fixed length prevents every element from falling independently. The equilibrium curve minimizes the total gravitational potential energy subject to the length constraint.

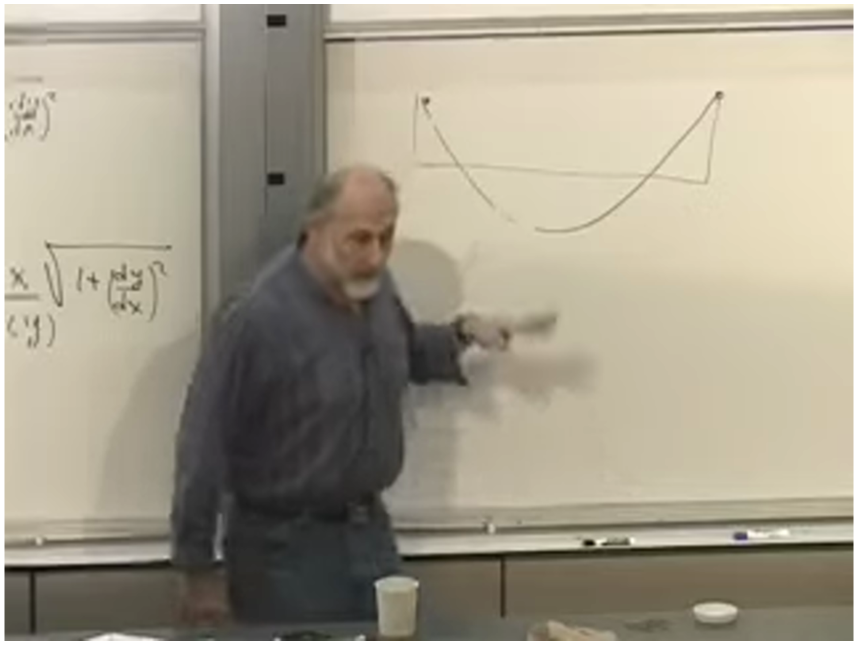

Figure 1.12: The hanging-chain sketch at 01:25:20. The fixed endpoints and fixed length constrain the chain while gravity lowers its center of mass.

Lecture frame: 01:25:20.

The resulting curve is a catenary, which may be written

$$y(x) = a \cosh\left(\frac{x - x_0}{a}\right) + y_0 \tag{1.42}$$

after the constants are chosen to fit the endpoints and length. Its derivation is not needed yet; its role is to show that minimizing a functional is a common physical task, not an artificial game invented for mechanics.

Question from the audience. From what height is the chain's potential energy measured? [a]

Response. Any fixed reference height will do. Shifting the zero adds the same constant to the energy of every allowed configuration, so it cannot change which configuration minimizes the energy.

[a] Lecture timestamp: 01:25:50–01:26:16.

These examples define the calculus of variations: find a function that makes a functional stationary. Mechanics will ask the same question for all the functions $q_i(t)$ that make up a trajectory.

1.7 The Action and the Lagrangian

We can now state the global formulation of mechanics. For every mechanical system there is a quantity called the action. Hold the endpoint configurations fixed and compare neighboring trajectories between them. The physical trajectory makes the action stationary.

Question from the audience. Is this really equivalent to Newton's formulation, or merely an approximation to it?[a]

Response. For systems governed by Newtonian mechanics, it is an equivalent formulation: varying the correct action yields Newton's equations. Which language is more efficient depends on the problem. The action principle also extends naturally to relativistic particles, fields, waves, and electromagnetism, where a literal catalogue of pairwise Newtonian forces is no longer the right description.

[a]Lecture timestamp: 01:28:21–01:30:13.

The practical advantage can be dramatic. Imagine describing a rigid eraser by writing force equations for every microscopic constituent. The local equations exist in principle, but they hide the small set of collective variables we actually care about: translation and rotation. A suitable action written in those variables reaches the same physics far more directly. Likewise, in a frictionless block-on-wedge problem, conservation laws or a Lagrangian can avoid pages of constraint forces. A more general principle is often not only deeper but faster.

For one particle moving in one dimension, the unknown path is $x(t)$. Since kinetic and potential energy have dominated the discussion, a natural first guess is

$$\int_{t_0}^{t_1} (T + U)\, dt. \tag{1.43}$$

That is the total energy integrated over time, and it is the wrong guess. The correct action is

$$A[x] = \int_{t_0}^{t_1} (T - U)\, dt \tag{1.44}$$

$$= \int_{t_0}^{t_1} \left[\frac{1}{2} m\dot{x}^2 - U(x)\right] dt. \tag{1.45}$$

The difference

$$L(x, \dot{x}) = T - U \tag{1.46}$$

is the Lagrangian, and the action is its integral over time.

The minus sign is not a decorative convention. If U is defined by $F = -dU/dx$, varying $T - U$ produces the correct sign in Newton's equation. One could rename $-U$ as a new quantity and write a plus sign, but that new quantity would then be the negative of potential energy. Renaming the symbol does not change the relation.

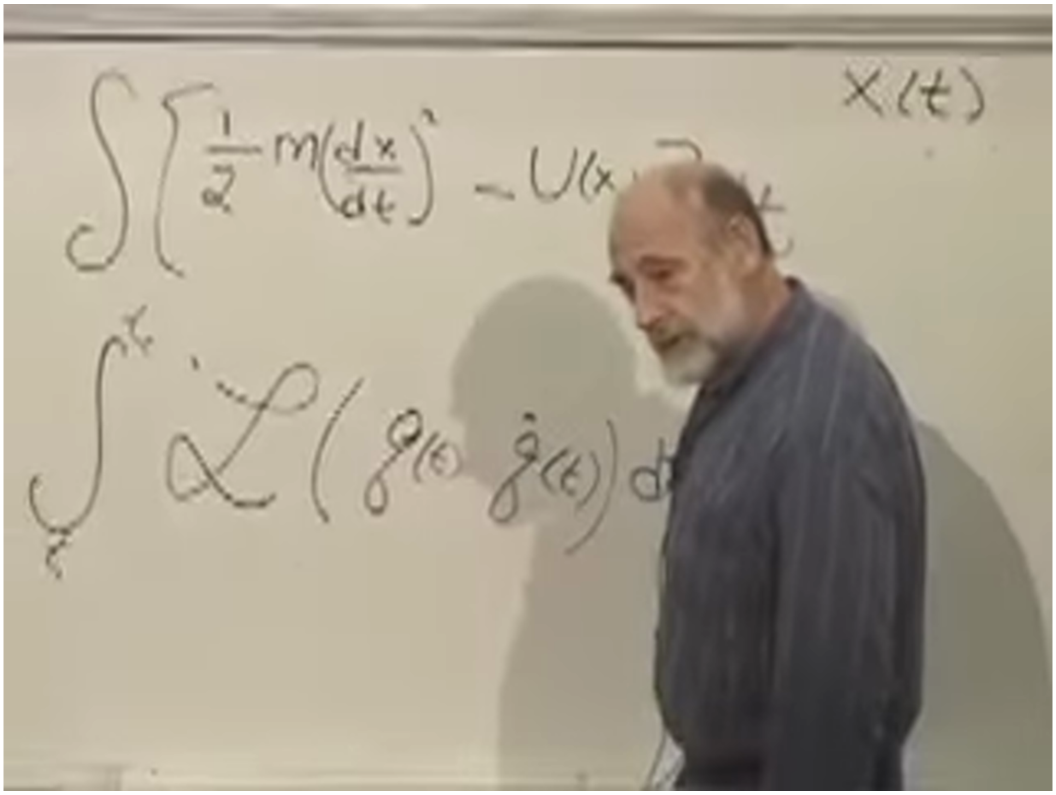

Figure 1.13: The action and its general Lagrangian form at 01:43:20. The independent variable is now time, and the path is the collection of functions $q_i(t)$.

Lecture frame: 01:43:20.

For generalized coordinates, write

$$L = L(\mathbf{q}, \dot{\mathbf{q}}), \qquad A[\mathbf{q}] = \int_{t_0}^{t_1} L(\mathbf{q}, \dot{\mathbf{q}})\, dt. \tag{1.47}$$

The analogy with the earlier examples is exact at the structural level. The arc-length integrand depends on y and y'; the optical integrand depends on y, y', and $c(y)$; the mechanical integrand depends on $\mathbf{q}$ and $\dot{\mathbf{q}}$. In every case an integral assigns a number to a whole path with fixed endpoints.

For ordinary Newtonian particles, $L = T - U$. More general classical systems can have less familiar Lagrangians, especially when velocity-dependent interactions or fields are present. What survives is the architecture: construct the action, vary the path while holding its endpoints fixed, and set the first-order change to zero.

Question from the audience. Why kinetic energy minus potential energy, rather than the total energy?[a]

Response. At this stage the honest test is operational. Use the calculus of variations and see which choice reproduces Newton's equations. $T - U$ does; $T + U$ gives the wrong force sign. The derivation, rather than an analogy with conserved energy, will explain why this is the right combination.

[a] Lecture timestamp: 01:33:30–01:36:38.

That derivation is the next task. Given a Lagrangian, we must learn how to turn stationary action into local equations of motion, and how to move back and forth between those equations and Newton's law. The same technique will solve the optical least-time problem. Mechanics and geometrical optics have arrived at the same mathematical threshold from different directions.